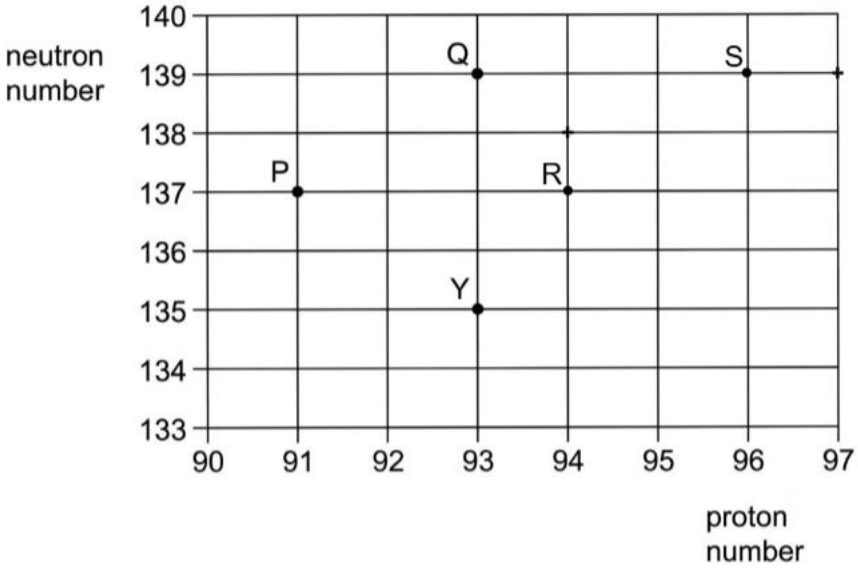
29 Nuclide Y is formed after two successive decays from another nuclide. Each decay can either be an alpha-emission or a beta-emission. Four other nuclides, P, Q, R and S are shown on the diagram.



Which of the nuclides P, Q, R and S is the initial nuclide that, after two successive decays, becomes Y?

A P **B** Q **C** R **D** S

Ans: A

An alpha decay is a transition 2 squares down and 2 squares to the left, since it is a reduction of 2 neutrons and 2 protons.

A beta decay is a transition of 1 square down and 1 square to the right, since the nucleus will have one less neutron and one more proton.

Working backwards:

P → Y through two beta decays

Q → Y requires two beta and one alpha decay, which is a total of three decays.

R and S cannot become Y through alpha and beta decays.